

8. Acquisition D: detection trials

A detection trial is one frame at a fresh random lateral offset. All features and blank sites sit on one plate, so a single frame yields one trial for every feature at once. The detection levels are the (feature, station) pairs: FEATURE_COUNT features at 9 stations give 27 values of the subtended size D_px, spaced by Z_STATION_RATIO, with each feature spanning two octaves and overlapping its neighbors by half an octave. The 10% and 0% minimums need many more trials at the levels where detection disappears, which lie at the far stations.

1. Post check. Run the quick-look detection count (Section 13, Step 2) on the first frame of each C pose at Z_REFERENCE_MM. If any post-only control site shows detections above the false-alarm rate, re-make the disk posts thinner and repeat the check.
2. Main trials. For each configuration in {disk, cutout} $\times$ {GAP_SMALL_MM, GAP_LARGE_MM} $\times$ every station of the Z ladder, capture DETECTION_TRIALS_PER_LEVEL poses. Each pose has a new random lateral offset, drawn as in Section 7, and gets FRAMES_PER_DETECTION_TRIAL frame.
3. Extended trials for the 0% point. At the DETECTION_ZERO_STATION_COUNT farthest stations, where the smallest feature lies below the expected threshold, increase the pose count to DETECTION_ZERO_TRIALS for every configuration. Every feature, and every blank site, then has that many trials there. With zero detections in n trials, the one-sided 95% upper bound on detection probability is about 3/n. So 300 trials bound a "never detected" level at $\leq 1\%$ (DETECTION_ZERO_PROBABILITY_BOUND).
4. Independence rule. Never use two frames from the same pose as separate trials. The fixed projector pattern and the target's fixed position make them strongly correlated. Section 13, Step 1 checks independence after the fact.
5. Reuse. The first frame of each C pose at a matching configuration may count as a D trial, which reduces the number of new poses needed.